

STRUCT SQUAD

Level Editor User Guide

Complete reference for the in-engine ImGui editor

Game Project GAM200 / GAM250 Engine

Group Struct Squad

Author GEYONGQI

Date 2026

Table of Contents

1. Overview
2. Getting Started
3. Interface Layout
4. Menu Bar
 - File Menu
 - Windows Menu
 - Editor Menu
5. Game Viewport
6. Entity Inspector
 - Search & Filter
 - Editing Components
 - Adding & Removing Components
 - Prefab Controls on an Entity
7. Entity Spawner
8. Asset Browser
9. Prefab Editor
10. Layer Visibility
11. Debug Info
12. Drag-and-Drop
13. Keyboard Shortcuts
14. Play, Pause & Stop
15. Tips & Common Mistakes

1 Overview

The **Level Editor** is a built-in tool powered by *Dear ImGui* that lets you inspect, create and modify game entities without recompiling the engine. It is only active in **Editor Mode** — once you click **PLAY**, the game runs normally and most editor panels are hidden or inactive.

Important

All changes made in Editor Mode (adding components, moving entities, saving a scene) are separate from changes that happen while the game is running. Stopping the simulation reverts to the last saved editor state.

2 Getting Started

- 1 Launch the game executable. The editor loads automatically in **Editor Mode**.
- 2 The window shows the **menu bar** at the top and several floating or docked panels.
- 3 Panel layouts are saved to `assets/imgui.ini`. You can drag panel title bars to re-dock them anywhere on screen.
- 4 To try the game, click **Editor** → **PLAY**. To return to editing, click **Editor** → **Stop**.
- 5 To save your work, use **File** → **Save Scene** (Ctr1 + S).

Warning

If you close the window without saving, all unsaved changes are lost. Always save before exiting.

3 Interface Layout

The editor is made up of several panels that can be freely docked and resized. Below is a description of every panel and where to find it.

Panel Name	What It Does	Default Visible
Game Viewport	Renders the game inside the editor window.	Yes
Entity Inspector	Lists every entity; edit components on the selected one.	Yes
Entity Spawner	Create blank entities; test audio.	Yes
Asset Browser	Browse project files; import assets; drag them into the scene.	Yes
Prefab Editor	Save/load/manage reusable entity templates (prefabs).	Yes
Layers	Show or hide individual render layers.	Yes
Debug Info	FPS, entity count, keyboard hint summary.	Yes
ImGui Demo	Built-in ImGui widget showcase (for developers).	No

All panels can be toggled on/off from the **Windows** menu.

4 Menu Bar

4.1 File Menu

Item	Shortcut	What It Does
Open Level...	<u>Ctrl</u> + <u>O</u>	Opens the Level Browser popup. Browse and double-click (or select + click Load) a Lua level script to load it into the editor.
Save Scene	<u>Ctrl</u> + <u>S</u>	Saves the current scene to assets/saves/<levelName>_save.json. The level name comes from the active game state.
Save Scene As...	—	Opens a text-input popup where you type a custom file path, then saves there.
Load Scene	—	Loads the scene JSON for the current level name from the default save path.
Load Scene...	—	Opens a text-input popup where you type the path to any .json scene file to load.
Current: <name>	—	Read-only label showing the name of the currently active level/game state.
Exit	—	Closes the application immediately.

Tip — Level Browser

In the Level Browser popup you can type in the

Search

box to filter levels by name. Rows prefixed with [F] indicate levels found on disk. Double-click a row to load it immediately.

4.2 Windows Menu

Each item in this menu is a toggle. Click once to show a panel; click again to hide it. You can also close any panel by clicking its × button.

Toggle	Panel it controls
Entity Inspector	The main component inspector.
Spawner	The Entity Spawner panel.
Debug Info	FPS and entity-count overlay.
ImGui Demo	Dear ImGui built-in demo window.
Assets	The Asset Browser.
Prefabs	The Prefab Editor.
Layers	The Layer Visibility panel.
Game Viewport	The in-editor game preview window.
Render to Viewport	When enabled, the game renders inside the Viewport panel instead of the main OS window.

4.3 Editor Menu

This menu controls the simulation state of the game.

Item	Available When	What It Does
PLAY	Simulation is stopped	Starts the game simulation. Editor panels become inactive.
Pause	Simulation is playing	Freezes all game logic without stopping the simulation.
Resume	Simulation is paused	Continues the paused simulation.
Stop	Simulation is playing or paused	Ends the simulation and returns to Editor Mode.
<i>Simulation: <state></i>	Always	Read-only status text showing <i>Stopped</i> , <i>Playing</i> , or <i>Paused</i> .

⚠ Warning

Any entities spawned or modified *while playing* are discarded when you click

Stop

. Only changes made in

Editor Mode

(before pressing PLAY) are persistent.

5 Game Viewport

The **Game** panel renders the game scene in real time inside the editor. It requires the panel to be at least **100 × 100 pixels** wide and tall; if it is too small, it displays a size warning instead.

How the Viewport works

- The game renders to an internal framebuffer (FBO) which is then displayed as a texture inside the ImGui window.
- The viewport is **resizable** — drag the panel edges to resize. The internal render resolution updates automatically.
- When **Render to Viewport** is on (Windows menu), the game draws inside this panel. When it is off, the game draws directly to the OS window behind the editor.
- Mouse clicks inside the Viewport while in **Editor Mode** are used for **entity picking** (see Section 6).

Note

The FPS Overlay (a small frame-time counter) is always shown in the corner of the editor, independent of the Debug Info panel.

6 Entity Inspector

The **Entity Inspector** is the most important panel for level editing. It lists every entity in the current scene on the left side and shows the **components** of the selected entity on the right.

Selecting an Entity

There are two ways to select an entity:

1. **Click in the list** — click any row in the Inspector to highlight it.
2. **Click in the Viewport** — in Editor Mode, left-click directly on an object in the Game Viewport. The entity under the cursor will be selected.

Note: only entities that have a **BoxCollider** or **CircleCollider** component can be picked this way. Entities without a collider cannot be clicked to select.

 Note

Grid tile entities are intentionally excluded from click-to-select to prevent accidentally selecting floor/wall tiles.

6.1 Search & Filter

At the top of the Inspector panel:

- **Total Entities** — shows how many entities exist in the scene.
- **Search by ID...** — type a number to find an entity by its numeric ID, or type part of a sprite name to find entities by sprite.
- **Clear** — resets the search box.
- **Show Filters** / **Hide Filters** — expands a panel of checkboxes to limit which entities appear in the list.

Filter checkboxes

When filters are shown, check the boxes for the component types you want to see. Only entities that have *all* checked component types will appear.

Filter	Shows entities that have...
Transform	a Transform component
Sprite	a Sprite component
MeshRenderer	a MeshRenderer component
Movement	a Movement component
BoxCollider	a BoxCollider component
CircleCollider	a CircleCollider component
Health	a Health component
AP	an Action Points component
SpriteAnimation	a SpriteAnimation component
AudioSource	an AudioSource component
ScriptComponent	a Lua Script component
Is Prefab Instance	entities that were spawned from a prefab

Click **Clear All Filters** to uncheck all boxes and show every entity again.

Pagination

If there are many entities, the list is split into pages. Use **< Prev** and **Next >** to navigate. The label "*Showing X – Y of Z*" tells you which entries are currently visible.

6.2 Editing Components

Click any entity row to select it. Its components appear as collapsible sections below. Expand a section by clicking its header. All editable fields update the entity immediately.

Transform

Field	Description
Position X / Y	World-space position of the entity.
Scale X / Y	Width and height scale multiplier.
Rotation	Rotation in degrees.

Sprite

Field	Description
Texture Path	File path to the image used for this sprite. Type a path directly or drag a texture from the Asset Browser.
Layer	Integer render layer. Higher numbers draw on top.
Tint	RGBA colour tint applied to the sprite (colour picker).

MeshRenderer

Field	Description
Sprite Name	Identifier name for this mesh's sprite sheet entry.
Layer	Render layer.
Order in Layer	Secondary sort within the same layer.
Tint	RGBA colour tint (colour picker).

Movement

Field	Description
Speed	Movement speed of the entity.
Direction X / Y	Current movement direction vector.

BoxCollider

Field	Description
Size X / Y	Width and height of the collision box.
Offset X / Y	Offset from the entity's transform origin.
Is Trigger	If checked, detects overlaps but does not block movement.

CircleCollider

Field	Description
Radius	Radius of the circle collider.
Offset X / Y	Offset from the entity's transform origin.

AudioSource

Field	Description
Sound Name	Name of the audio clip (must match a loaded audio asset).
Volume	Playback volume (0.0 – 1.0).
Pitch	Playback pitch multiplier.
Loop	If checked, the sound repeats continuously.
Play (on start)	If checked, the sound plays automatically when the scene starts.

ScriptComponent (Lua Script)

Field / Button	Description
Script Path	File path to the .lua script attached to this entity.
Browse...	Opens the Script Browser popup to search and select a .lua file from assets/scripts/.
Lua State	Read-only indicator of the script's current runtime state.
OnInit / OnUpdate / OnDestroy	Read-only indicators showing which lifecycle callbacks exist in the script.
Update Timer	Shows the average time (ms) spent running this script per frame.

Health

Field	Description
Max Health	Maximum HP value.
Current Health	Current HP value.
Is Dead	Read-only indicator — shown greyed-out when HP is 0.

AP (Action Points)

Field	Description
Action Points	Current AP value.
Max Action Points	Maximum AP value.

AttackAP

Field	Description
Attack Points	Current attack AP value.
Max Attack Points	Maximum attack AP value.

AttackRangeComponent

Field	Description
Min Range	Minimum attack range.
Max Range	Maximum attack range.
Show Range	If checked, shows a visual range indicator in the game view.

Global Settings (always visible at the top of the Inspector)

- **Master Volume** slider — adjusts the global audio volume for the entire game immediately.

6.3 Adding & Removing Components

Adding a Component

- 1 Select an entity in the Inspector list.
- 2 Scroll down to the bottom of its component list.
- 3 Click the **Add Component** button. A popup menu appears.
- 4 Click the component type you want to add. It appears immediately on the entity.

The following component types can be added this way:

- Transform
- Movement
- Sprite
- MeshRenderer
- BoxCollider
- CircleCollider
- AudioSource
- ScriptComponent
- Health
- SpriteAnimation
- AP (Action Points)
- AttackRangeComponent

- Chest
- Goal
- Inventory
- AttackAP

Note

An option only appears in the Add Component popup if the entity does *not already have* that component.

Removing a Component

Every component section has a **Remove** button at its bottom-right. Click it to delete that component from the entity. This action cannot be undone unless you use Ctrl + Z immediately after.

6.4 Prefab Controls on an Entity

If the selected entity was created from a prefab, a green strip appears at the top of its component list showing the prefab file name. From here you can:

Button	What It Does
Revert to Prefab	Discards all changes and restores the entity to the saved prefab definition.
Delete	Removes the entity from the scene entirely.
Save Prefab	Overwrites the original prefab file with this entity's current state.
Save as New Prefab	Saves as a brand-new prefab file, keeping the original unchanged.
Duplicate	Creates a copy of this entity in the scene at the same position.

7 Entity Spawner

The **Entity Spawner** panel lets you add new blank entities to the scene and test audio playback.

Spawning a blank entity

- 1 Set **Spawn X** and **Spawn Y** drag-fields to the world position where you want the entity to appear.
- 2 Click **Spawn Blank Entity** . A new entity is created with a **Transform** component at that position.
- 3 Select the new entity in the Inspector and add the components you need.

Audio testing buttons

Button	What It Does
Trigger Menu BGM	Plays the menu background music track (mmbgm).
Trigger In-Game BGM	Plays the in-game background music track (igbgm).
Stop All Audio	Stops all currently playing audio immediately.

Drop zones

- **Drop Audio Here** — drag an audio asset from the Asset Browser onto this zone to test it.
- **Drop Prefab Here** — drag a .prefab file from the Asset Browser onto this zone to spawn it at the current Spawn X/Y position.

8 Asset Browser

The **Asset Browser** lets you navigate the project's file system, import new assets, and use files directly in the scene.

Navigation

- The current folder path is shown at the top as a bold label.
- Entries beginning with -> are **folders**. Click a folder to open it.
- Click the < button (top-left) to go up one folder level.

File actions by type

File Type	Single-click	Double-click	Drag
Image (.png, .jpg)	—	Spawns a new sprite entity at world origin using this texture.	Drag with payload Sprite → drop onto a Sprite or MeshRenderer field in the Inspector to assign the texture.
Prefab (.prefab)	—	Loads and spawns the prefab into the scene.	Drag with payload Prefab → drop onto the Spawner's <i>Drop Prefab Here</i> zone.
Scene (.json)	—	Loads the JSON scene file, replacing the current scene.	—
Level script (.lua with "level" or "menu" in the name)	—	Clears the scene and loads this Lua level script in editor mode.	—
Text level (.txt)	—	Loads the level from this text file (clears current scene).	—
Any other file	—	No action.	—

Importing an asset

- 1 Navigate to the folder where you want the asset to live.
- 2 Click **Import Asset...** . A Windows file picker dialog opens.
- 3 Select any file on your computer. The file is copied into the current browser folder.
- 4 Audio files (.wav) open an additional **Import Audio Asset** popup to confirm settings. Unsupported audio formats show an error popup instead.
- 5 Click **Refresh** if the imported file does not appear immediately.

Deleting a file

- 1 Right-click any **file** (not a folder) in the browser.
- 2 Click **Delete...** in the context menu.
- 3 A confirmation popup appears: click **Delete** to permanently remove the file from disk, or **Cancel** to abort.

Warning

Deleting a file through the Asset Browser removes it *permanently from disk*. There is no undo for this action.

9 Prefab Editor

A **prefab** is a saved template of an entity (its components and values) that can be reused and spawned multiple times. The Prefab Editor panel manages all prefab operations.

Saving an entity as a prefab

- 1 Under "**Save Entity as Prefab**", use the **Select Entity** dropdown to choose which entity to save.
- 2 Type a name in the **Prefab Name** field.
- 3 Click **Save as Prefab** . The prefab is saved to assets/prefabs/.

Loading a prefab into the scene

- 1 Under "**Load Prefab**", click **Refresh List** to update the list.
- 2 Click a prefab name in the **Available Prefabs** list to select it.
- 3 Set the **Spawn Position** X / Y values.
- 4 Click **Load Prefab** . The entity appears in the scene.

Prefab Instances

The **Prefab Instances** section shows a count and a read-only list of every entity in the scene that was spawned from a prefab.

Prefab Inspector (bulk edit)

This section shows the **Transform, Sprite, MeshRenderer, Movement, BoxCollider, and CircleCollider** fields of the *first* prefab instance. Changes here are **shared across all instances** of that prefab.

Cross-Scene Update

- **Apply Prefab to ALL Scenes** — saves the current prefab definition and applies it to every scene file that uses it.

- **Clear All Tracking** — removes all instance-tracking data for prefabs in the current session.

10 Layer Visibility

The **Layers** panel lets you hide or show groups of objects by their render layer. This is useful for focusing on specific parts of the scene while editing. It only affects the **editor view**; hidden layers are still visible in-game.

Note

This panel is only meaningful in **Editor Mode**

. Layer visibility has no effect on the game simulation.

Layer Name	Layer Number	Typical Contents
Background	-1000	Sky, background scenery
Ground	0	Floor tiles, terrain
Props / Items	1	Interactable objects, chests
Enemies / Characters	2	Enemy and player sprites
Range Indicators	3	Attack range visuals
Player	4	Player character overlay
Projectiles	5	Arrows, spells
Effects	10	Particle effects, flashes
UI	100	HUD, health bars, buttons
Overlay	1000	Screen overlays, transitions

- Click **Show All** to make all layers visible again.
- At the bottom, use the **Custom Layer** number field + **Visible** checkbox to hide/show any layer by its exact number.

11 Debug Info

The **Debug Info** panel shows real-time performance data and a quick reference of editor keyboard controls.

Displayed Value	Meaning
ms/frame	Time each frame takes to compute, in milliseconds.
FPS	Frames per second.
Entities	Total number of entities currently in the scene.

The panel also lists the following **editor camera controls** as a hint:

Key	Action
	Move the editor camera.
	Zoom in.
	Zoom out.
	Reset camera to default position.
	Jump to Level 3 (game state shortcut).
	Jump to Main Menu (game state shortcut).
	Quit the application.
Editor → PLAY	Enter Play Mode.

Note

An always-visible

FPS Overlay

is also drawn in the corner of the editor window. Its colour changes by frame rate: green = good, yellow = moderate, red = slow.

12 Drag-and-Drop

The editor supports two types of drag-and-drop: **within the editor** (ImGui panels) and **from the OS** (Windows Explorer).

12.1 Drag from Asset Browser (inside the editor)

What you drag	Where to drop	Result
Image file (.png, .jpg)	Sprite / MeshRenderer texture field in the Inspector	Assigns that image as the texture for that component.
Prefab file (.prefab)	"Drop Prefab Here" zone in the Spawner	Spawns the prefab at the Spawner's set position.

12.2 Drag from Windows Explorer (OS drag-and-drop)

You can drag files directly from Windows Explorer onto the game window:

File type	What happens
.txt (level file)	The current scene is cleared and the level described in the text file is loaded.
.png, .jpg, .jpeg (image)	If the mouse is over the Game Viewport, a new sprite entity is spawned at the drop position.
.wav (audio)	Opens the Import Audio Asset popup to import the file.
Unsupported audio format	Shows an Audio Format Error popup with the list of supported formats.
.prefab / .json	Loads the prefab or scene into the editor.
.lua (level-named script)	Clears the scene and loads the Lua level in editor mode.
.lua (non-level script) + mouse over Viewport	Creates a new entity at that position with Transform, a default sprite, a BoxCollider (trigger), and the dragged script attached.

13 Keyboard Shortcuts

Shortcut	Action
<u>Ctrl</u> + <u>O</u>	Open Level (Level Browser popup).
<u>Ctrl</u> + <u>S</u>	Save Scene to JSON.
<u>Ctrl</u> + <u>Z</u>	Undo last action.
<u>Ctrl</u> + <u>Shift</u> + <u>Z</u>	Redo last undone action.
<u>Ctrl</u> + <u>Y</u>	Redo (alternative shortcut).
<u>Ctrl</u> + <u>C</u>	Copy the selected entity.
<u>Ctrl</u> + <u>V</u>	Paste the copied entity into the scene.
<u>Delete</u>	Delete the selected entity (Editor Mode only).
<u>↑ ↓ ← →</u>	Move the editor camera.
<u>1</u>	Zoom in the editor camera.
<u>2</u>	Zoom out the editor camera.
<u>0</u>	Reset the editor camera position.
<u>3</u>	Load Level 3 game state.
<u>5</u>	Load Main Menu game state.
<u>Q</u> or <u>ESC</u>	Quit the application.

Note

Shortcuts Ctrl + Z , Ctrl + C , Ctrl + V and Delete only work when no text input box is currently focused. Click somewhere neutral (e.g. the viewport) first if a shortcut is not responding.

14 Play, Pause & Stop

Action	Where	Effect
PLAY	Editor menu → PLAY	Starts the game simulation. The editor switches to Play Mode; most editing features are disabled. The camera automatically follows the player entity.
Pause	Editor menu → Pause (while playing)	Freezes the simulation. No logic runs but the window stays open. The status label changes to <i>Paused</i> .
Resume	Editor menu → Resume (while paused)	Continues the paused simulation from where it left off.
Stop	Editor menu → Stop (while playing or paused)	Ends the simulation and returns to Editor Mode. The scene resets to its pre-play state. The editor camera control returns.

Simulation Status

The current simulation state is always shown at the bottom of the Editor menu as: "*Simulation: Stopped*", "*Simulation: Playing*", or "*Simulation: Paused*".

15 Tips & Common Mistakes

Tips for beginners

- **Save often.** Use `Ctrl + S` every few minutes, especially before pressing PLAY.
- **Docking panels.** Drag any panel's title bar over another panel or the edge of the window to dock it. This keeps your workspace tidy.
- **Entity not clickable?** You can only click-to-select entities that have a BoxCollider or CircleCollider. Add one if you need click-to-select.
- **Can't find an entity?** Use the Search by ID field in the Inspector, or enable the component filter to narrow the list.
- **Texture not showing?** Make sure the file path in the Sprite or MeshRenderer texture field is correct and that the file exists in assets/.
- **Script not running?** Check the ScriptComponent path is correct. The "Initialized" indicator in the Inspector must appear for the script to be active.
- **Layer order.** If an object appears behind or in front of the wrong things, check its render layer number. Higher numbers appear on top.

Common mistakes

Mistake	How to fix it
Making changes while the game is running, then wondering why they disappeared after Stop.	Always switch back to Editor Mode (Stop) before editing. Changes during Play are temporary.
Saving a scene but forgetting to save after adding more entities.	Use <u>Ctrl</u> + <u>S</u> frequently. Check the title bar for unsaved state.
Deleting a file in the Asset Browser and breaking a scene reference.	Only delete files you are certain are not used by any entity, script, or scene.
Dragging an image from Explorer but the sprite appears at world (0, 0) instead of where you dropped it.	Make sure the mouse is hovering over the Game Viewport (not just anywhere on the screen) when you release the file.
Entity invisible in the editor but visible in-game.	Check the Layers panel — the entity's render layer may be hidden in the editor.
Undo not working.	Click somewhere neutral (not in a text field) then press <u>Ctrl</u> + <u>Z</u> .